

Multiple Choice

1. **Answer:** A

Options: A) $p(y|x, w)$; B) $p(y, x)$; C) $p(w|x, w)$; D) None of the above

Note: Correct option: A - $p(y|x, w)$

2. **Answer:** C

Options: A) $P(A | B) * P(B | C) * P(C | A)$; B) $P(C | A, B) * P(A) * P(B)$; C) $P(A, B | C) * P(C)$; D) $P(A | B, C) * P(B | A, C) * P(C | A, B)$

Note: Correct option: C - $P(A, B | C) * P(C)$

3. **Answer:** D

Options: A) Increase bias; B) Decrease bias; C) Increase variance; D) Decrease variance

Note: Correct option: D - Decrease variance

4. **Answer:** D

Options: A) L 0 norm; B) L 1 norm; C) L 2 norm; D) either (a) or (b)

Note: Correct option: D - either (a) or (b)

5. **Answer:** B

Options: A) $O(1)$; B) $O(N)$; C) $O(\log N)$; D) $O(N^2)$

Note: Correct option: B - $O(N)$

True / False

6. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Forward stage wise selection'.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Lasso'.

8. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Detecting fraudulent credit card transactions'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** `def find_demlo(s): | return res`

Note: Students should provide a detailed explanation referencing algorithm steps.

12. **Answer:** `def sumofFactors(n) : | return 0 | return res`

Note: Students should provide a detailed explanation referencing algorithm steps.

End of Answer Key